# Did Liquidity Govern Stablecoin Peg Pass-Through? Evidence from the March 2023 USDC Depeg

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February 2026, Revised in October 2026

# Stablecoin depegs and arbitrage

- A stablecoin is a cryptocurrency that trades at a fixed target against a fiat currency, one US dollar (USD) for Tether (USDT) and USD Coin (USDC). The stablecoin price of Bitcoin (BTC) is a dollar price only while the peg holds.
- Perez Riaza and Gnabo (2025) find that a Tether depeg raises the probability of a jump in BTC/USD almost fivefold within 5 minutes. Bartůněk (2024) finds that depegging raises the realized volatility of Bitcoin.
- Makarov and Schoar (2020) document large arbitrage deviations across crypto exchanges. Reynolds et al. (2021) find deviations from triangular arbitrage parity in Bitcoin markets but not among fiat currencies.
- Lyons and Viswanath-Natraj (2023) show that Tether's peg deviations halved once arbitrage with the issuer became easier.

# Research question

## Question

Did liquidity govern how stablecoin peg stress passed into the stablecoin price of Bitcoin during the March 2023 USDC depeg?

## Finding

1. During the crisis, the peg deviation carried most of the basis variance, and the triangular arbitrage deviation had a half-life below 1 minute.
2. Liquidity did not measurably govern the pass-through.
3. The effect of USDC/USD illiquidity is significant at 1-minute sampling but not robust.
4. The stress that reached the stablecoin price of Bitcoin was the peg itself, that arbitrage relationships were maintined.

# Sample and notation

$$F_t = \mathbf{1}\{v_t^{\mathrm{BTC/USD}} > 0,\ v_t^{\mathrm{BTC}/Q} > 0,\ v_t^{Q/\mathrm{USD}} > 0\}, \qquad FF_t = F_t F_{t-1}$$

- We use 1-minute data from Binance.US for five pairs, BTC/USD, BTC/USDT, BTC/USDC, USDT/USD and USDC/USD, from 1 to 21 March 2023.
- Write $P_t^{X/Y}$ for the close price of one unit of X in units of Y at time $t$, and $Q \in \{\mathrm{USDT}, \mathrm{USDC}\}$ for a stablecoin.
- Deviations are in basis points (bps), so log prices are scaled by $m = 10^4$.
- $F_t$ therefore indicates that all three pairs of the triangle trade at time $t$, and $FF_t$ that both $t$ and $t-1$ have $F = 1$, with $v$ the pair volume.

# Depeg event identification

- Following Perez Riaza and Gnabo (2025), $\pi_k$ is the 5-minute $Q$/USD close.
- A bar is a breach when $|\pi_k - 1| \geq \tau$, with $\tau = 0.5\%$ in the baseline, as in Perez Riaza and Gnabo (2025) and Polizu et al. (2023).
- An event starts at the last bar before the breach when the price equals 1 dollar exactly, or at the first breach bar when the previous bar is at the peg.
- Estimates for one stablecoin use its own window, and estimates that use all pairs use the union of the two windows, which equals the USDT window.

| Stablecoin | Threshold and start | Start | End | Hours |
|---|---|---|---|---|
| USDT | ±0.5%, last-peg start (baseline) | 10 Mar 21:10 | 15 Mar 13:10 | 112.0 |
| USDC | ±0.5%, last-peg start (baseline) | 11 Mar 01:20 | 14 Mar 02:30 | 73.2 |

Depeg windows from the 5-minute $Q$/USD close. Times are UTC in March 2023.

# Liquidity measures

$$\text{Kyle}_t = m \left( \frac{\frac{1}{5} \sum_{i=0}^{4} r_{t-i}^2}{\sum_{i=0}^{4} V_{t-i}} \right)^{1/3}$$

- We measure illiquidity with the Kyle estimator of Kyle and Obizhaeva (2016), where $r_t$ is the 1-minute log return and $V_t$ the quote volume of a pair.
- $\text{Kyle}_t$ rises with volatility per unit of volume.

# Liquidity by pair and regime

| | Kyle | | | Quote volume per day | | |
|---|---|---|---|---|---|---|
| Pair | pre | crisis | post | pre | crisis | post |
| BTC/USD | 0.79 | 1.02 | 0.88 | 141 | 254 | 333 |
| BTC/USDT | 1.14 | 1.40 | 1.73 | 55.4 | 113 | 58.1 |
| BTC/USDC | 3.08 | 6.37 | 5.33 | 4.15 | 5.38 | 3.71 |
| USDT/USD | 0.91 | 1.48 | 1.44 | 3.21 | 8.99 | 3.89 |
| USDC/USD | 1.23 | 7.23 | 3.79 | 1.13 | 9.76 | 0.94 |

Liquidity by pair and regime over the union window. Quote volume is in millions of the quote currency.

- The crisis affected the USDC pairs most.
- The Kyle illiquidity of USDC/USD rose $1.23 \to 7.23$ and that of BTC/USDC $3.08 \to 6.37$, compared with $0.79 \to 1.02$ for BTC/USD.

# Basis decomposition

$$D_t = m\left(\ln P_t^{\mathrm{BTC}/Q} - \ln P_t^{\mathrm{BTC/USD}}\right), \qquad \delta_t = m \ln P_t^{Q/\mathrm{USD}}, \qquad b_t = -D_t - \delta_t$$

- For stablecoin $Q$, we decompose the cross-currency basis $D_t$ into the peg deviation $\delta_t$ and the triangular arbitrage deviation $b_t$.
- Triangular arbitrage parity, $P^{\mathrm{BTC/USD}} = P^{\mathrm{BTC}/Q} P^{Q/\mathrm{USD}}$, implies $b_t = 0$ (Reynolds et al., 2021), so full pass-through of the peg means $D_t = -\delta_t$.
- $b_t$ is the part of the basis that the peg does not explain.
- The variance of the basis decomposes into

$$\mathrm{Var}(D) = \mathrm{Var}(\delta) + \mathrm{Var}(b) + 2\,\mathrm{Cov}(\delta, b).$$

# Half-life of deviations

$$x_t = c + \rho x_{t-1} + u_t, \qquad \text{HL} = \frac{\ln 0.5}{\ln \rho}$$

- For $x \in \{\delta, b\}$, we estimate a first-order autoregressive model and the half-life HL in minutes (Lyons and Viswanath-Natraj, 2023).

# Pass-through regression

$$\Delta b_t = \alpha + \beta b_{t-1} + \left(\lambda_0 + \boldsymbol{\lambda}_1^\top \tilde{\mathbf{z}}_{t-1}\right) \delta_{t-1} + \boldsymbol{\theta}^\top \tilde{\mathbf{z}}_{t-1} + \kappa \widetilde{\mathrm{RV}}_{t-1} + \varepsilon_t$$

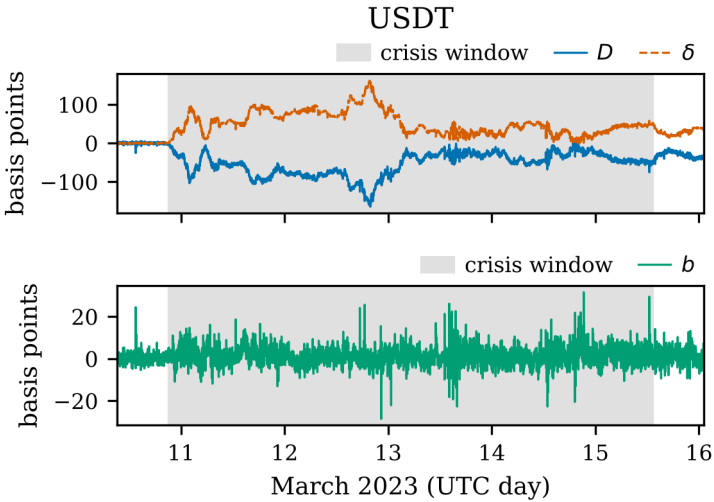
- CS is the spread estimator of Corwin and Schultz (2012), RV the realized volatility (Appendix A.1), and a tilde marks standardization within the estimation sample.
- The vector $\mathbf{z}_{t-1}$ holds the lagged CS and Kyle illiquidity of BTC/$Q$ and the lagged Kyle illiquidity of $Q$/USD, and $\mathrm{RV}_{t-1}$ is the lagged RV of BTC/USD.
- $\boldsymbol{\theta}$ and $\kappa$ are control coefficients, $\beta$ measures the reversion speed of $b$, and $\lambda_0$ measures how the lagged peg deviation moves $b$ at average liquidity.
- The vector $\boldsymbol{\lambda}_1$ measures how that response changes with a 1 standard deviation rise in each illiquidity measure, so liquidity governs the pass-through when $\boldsymbol{\lambda}_1 \neq \mathbf{0}$.

# Spillover index and Granger causality

$$\mathbf{r}_t = \mathbf{c} + \sum_{l=1}^{p} \mathbf{\Phi}_l \mathbf{r}_{t-l} + \boldsymbol{\varepsilon}_t, \qquad \mathrm{E}\left[\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_t^{\top}\right] = \mathbf{\Sigma}, \qquad S = \frac{1}{5} \sum_{i \neq j} \tilde{\phi}_{ij}$$

- We model the joint dynamics of the five pairs by a vector autoregression (VAR) on 5-minute scaled log returns, with the lag order $p$ is chosen by the BIC up to 6 lags.
- $\phi_{ij}(H)$ is the generalized share of the $H$-step forecast error variance of pair $i$ due to shocks in pair $j$ (Koop et al., 1996, Pesaran and Shin, 1998).
- The normalized shares $\tilde{\phi}_{ij}$ sum to 100 in each row (Appendix C.1), and $S$ is the total spillover index of Diebold and Yilmaz (2012) with $H = 10$ (50 minutes).
- Pair $j$ Granger-causes pair $i$ (Granger, 1969) when the lags of $j$ help predict $i$, which we test with a Wald test that all $p$ coefficients of $j$ in the equation of $i$ are zero.

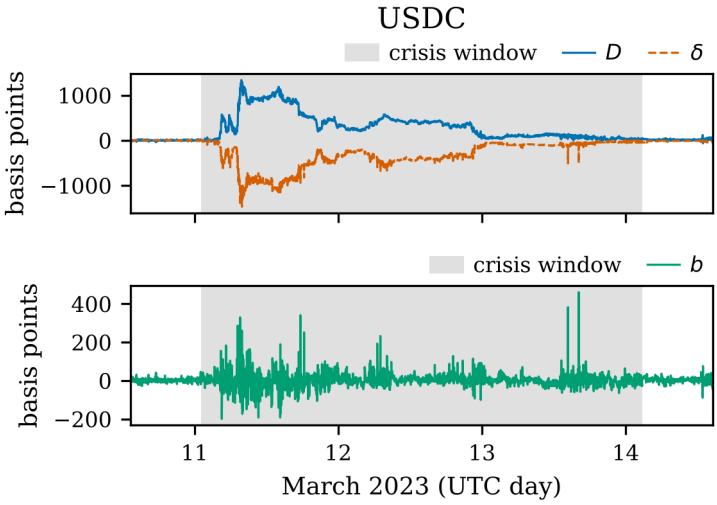
# Decomposition of the basis, USDT




- Before the depeg, all three series stayed close to zero.
- Once the peg broke, $D$ mirrored $\delta$ almost exactly, while $b$ fluctuated around zero with short spikes.

Cross-currency basis $D$ and peg deviation $\delta$ (top), and triangular arbitrage deviation $b$ (bottom) in bps, 1-minute close prices from 12 hours before to 12 hours after each crisis window (shaded), UTC.

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# Decomposition on synchronous minutes

| | USDT | | | USDC | | |
|---|---|---|---|---|---|---|
| | pre | crisis | post | pre | crisis | post |
| Mean $D$ | $-0.11$ | $-53.2$ | $-26.0$ | 0.13 | 399 | 7.23 |
| Mean $\delta$ | 0.12 | 51.6 | 25.3 | $-0.61$ | $-400$ | $-7.17$ |
| $\mathrm{Var}(\delta)/\mathrm{Var}(D)$ | 0.56 | 1.02 | 0.97 | 0.67 | 1.00 | 0.63 |
| Mean $b$ | $-0.01$ | 1.57 | 0.70 | 0.47 | 1.72 | $-0.06$ |
| $\lvert b\rvert$ median | 1.03 | 2.50 | 2.39 | 2.23 | 13.4 | 5.10 |
| $\lvert b\rvert$ 99th percentile | 5.37 | 12.3 | 10.9 | 17.3 | 140 | 26.5 |

Decomposition on synchronous minutes ($F_t = 1$), in bps. The remaining moments are in Appendix B.1.

- During the crisis, $\mathrm{corr}(D, -\delta) = 0.99$ for both stablecoins and $\mathrm{Var}(\delta)/\mathrm{Var}(D)$ is 1.02 for USDT and 1.00 for USDC, so $\delta$ accounts for almost the entire variance of $D$.
- The crisis widened the distribution of $b$, whose mean was below 2 bps, while for USDC the median of $|b|$ rose $2.23 \to 13.4$ bps.

# Half-lives

| | | pre | crisis | post |
|---|---|---|---|---|
| USDT | $\delta$ | 5.72 | 158 | 27.8 |
| | $b$ | 0.45 | 0.82 | 0.64 |
| USDC | $\delta$ | 0.92 | 63.9 | 2.47 |
| | $b$ | 0.43 | 0.52 | 0.61 |

Half-life in minutes on consecutive synchronous minutes ($FF_t = 1$).

- During the crisis, the half-life of $b$ is 0.82 minutes for USDT and 0.52 minutes for USDC, compared with 158 and 63.9 minutes for $\delta$.
- Arbitrage therefore halved triangular deviations within a minute even at the height of the stress, while the peg deviation persisted for hours.
- The persistence of the basis was hence the persistence of the peg.

# Liquidity and pass-through

| | USDT | | | USDC | | |
|---|---|---|---|---|---|---|
| | pre | crisis | post | pre | crisis | post |
| $\beta$ | $-0.810^{***}$ | $-0.589^{***}$ | $-0.688^{***}$ | $-0.898^{***}$ | $-0.744^{***}$ | $-0.691^{***}$ |
| | $(-13.77)$ | $(-18.92)$ | $(-23.47)$ | $(-13.42)$ | $(-12.84)$ | $(-7.88)$ |
| $\lambda_0$ | $-0.0782^{***}$ | $-0.0098^{***}$ | $-0.0387^{**}$ | $-0.7004^{***}$ | 0.0015 | 0.0780 |
| | $(-2.82)$ | $(-2.88)$ | $(-2.53)$ | $(-7.67)$ | (0.24) | (1.26) |
| $\lambda_1$, CS of BTC/$Q$ | 0.0161 | $-0.0019$ | $-0.0096^{**}$ | $0.2642^{***}$ | $-0.0006$ | $-0.0079$ |
| | (0.61) | $(-1.02)$ | $(-2.09)$ | (2.63) | $(-0.13)$ | $(-0.24)$ |
| $\lambda_1$, Kyle of BTC/$Q$ | 0.0124 | $-0.0045^{*}$ | $0.0205^{**}$ | 0.0735 | 0.0027 | $-0.0220$ |
| | (0.34) | $(-1.65)$ | (2.20) | (1.08) | (1.06) | $(-1.25)$ |
| $\lambda_1$, Kyle of $Q$/USD | $0.0301^{*}$ | $-0.0009$ | $-0.0060$ | $0.1195^{***}$ | $0.0137^{***}$ | $-0.0317$ |
| | (1.89) | $(-0.31)$ | $(-1.01)$ | (7.52) | (2.91) | $(-1.06)$ |
| Joint $p$, $\lambda_1 = 0$ | 0.208 | 0.263 | 0.021 | $<0.001$ | 0.007 | 0.568 |

Pass-through regression on consecutive synchronous minutes, with $t$ statistics in parentheses.
$^{*}$, $^{**}$ and $^{***}$ denote significance at 10%, 5% and 1%.

- During the crisis, the liquidity terms are jointly significant for USDC but not for USDT.
- At the mean discount of $-400$ bps, a 1 standard deviation rise in the Kyle illiquidity of USDC/USD implies a move in $b$ of about $-5.5$ bps per minute.

# Variance shares

| | pre | crisis | post |
|---|---|---|---|
| Panel A. Variance shares (%) | | | |
| Share of BTC/USDC from BTC/USD | 32.4 | 21.4 | 31.1 |
| Share of BTC/USDC from USDC/USD | 0.0 | 5.9 | 0.2 |
| Share of USDC/USD from BTC/USDC | 0.2 | 10.3 | 0.3 |

Spillovers of 5-minute returns over the union window, Panel A. Shares are the normalized $\tilde{\phi}_{ij}$ with $H = 10$.

- The total spillover index barely moved, at $39.7 \to 37.5 \to 39.4$.
- The crisis did not change overall connectedness but did change its composition.
- The share of BTC/USDC forecast error variance due to BTC/USD fell from 32.4% before the crisis to 21.4% during it, and USDC/USD took part of its place.

# Granger causality

| | pre | crisis | post |
|---|---|---|---|
| Panel B. Granger causality, Wald statistic | | | |
| BTC/USD → BTC/USDC | 17.79*** | 10.11*** | 28.77*** |
| BTC/USDC → BTC/USD | 3.51 | 1.89 | 7.54 |
| USDC/USD → BTC/USDC | 1.37 | 14.87*** | 5.27 |
| BTC/USDC → USDC/USD | 6.00 | 114.19*** | 7.06 |
| BTC/USD → BTC/USDT | 16.76** | 6.99** | 29.54*** |
| BTC/USDT → BTC/USD | 10.06 | 13.57*** | 3.05 |
| BTC/USDT → USDT/USD | 28.45*** | 15.22*** | 28.42*** |
| USDT/USD → BTC/USDT | 5.15 | 0.03 | 4.84 |

Spillovers of 5-minute returns over the union window, Panel B. *, ** and *** denote significance at 10%, 5% and 1%.

- BTC/USD led BTC/USDC in every regime and never the reverse.
- USDC/USD and BTC/USDC led each other only during the crisis.
- During the crisis, BTC/USDC partly became a market for USDC itself, so that its prices and the USDC peg predicted each other.

# Robustness of key crisis estimates

| | USDC pass-through | | | Half-life, USDT / USDC | | |
|---|---|---|---|---|---|---|
| Check | $\lambda_1$ | Joint $p$ | USDT $\lambda_0$ | $b$ | $\delta$ | Share |
| R0 baseline | $0.0137^{***}$ (2.91) | 0.007 | $-0.0098^{***}$ ($-2.88$) | 0.82 / 0.52 | 158 / 63.9 | 21.4 |
| R1 5-minute bars | $0.0143^{*}$ (1.83) | 0.106 | $-0.0132^{***}$ ($-3.16$) | 2.47 / 2.27 | 284 / 219 | – |
| R2 CS of BTC/USDC | 0.0005 (0.11) | 0.910 | $-0.0084^{**}$ ($-2.36$) | – | – | – |
| R2 Kyle of BTC/USDC | 0.0040 (1.33) | 0.185 | $-0.0091^{**}$ ($-2.56$) | – | – | – |
| R2 Kyle of USDC/USD | $0.0149^{***}$ (2.93) | 0.003 | $-0.0093^{***}$ ($-2.70$) | – | – | – |

Key crisis estimates, with $t$ statistics in parentheses. *, ** and *** denote significance at 10%, 5% and 1%.

- R1 re-estimates the half-lives and the pass-through regression on 5-minute bars, and R2 enters each liquidity measure alone.
- On 5-minute bars the coefficient on the Kyle illiquidity of USDC/USD stays similar but is significant only at 10%, with a joint $p$ of 0.106.
- The effect of USDC/USD illiquidity is significant at 1-minute sampling but not robust.

# Conclusion

- We decomposed the cross-currency basis of USDT and USDC on 1-minute data and examined whether the response of $b$ to $\delta$ depended on lagged illiquidity.
- During the crisis, $\delta$ carried almost the entire variance of $D$, and $b$ had a half-life below 1 minute, against about 1 hour for the USDC peg and over 2.5 hours for the USDT peg.
- Liquidity did not measurably govern the pass-through. The effect of USDC/USD illiquidity is significant at 1-minute sampling but not robust.
- The stress that reached the stablecoin price of Bitcoin was the peg itself. During the crisis, BTC/USDC partly decoupled from BTC/USD and traded with the USDC peg.

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Code and replication data are available at *GitHub*.

# Appendix A.1. Liquidity and volatility measures

Write $\overline{P}_t$ and $\underline{P}_t$ for the high and the low of a bar. The spread estimator of Corwin and Schultz (2012) over bars $t-1$ and $t$ is defined by

$$\psi_t = \sum_{i=0}^{1} \left( \ln \frac{\overline{P}_{t-i}}{\underline{P}_{t-i}} \right)^2, \qquad \omega_t = \left( \ln \frac{\max(\overline{P}_t, \overline{P}_{t-1})}{\min(\underline{P}_t, \underline{P}_{t-1})} \right)^2,$$

$$\zeta_t = \frac{\sqrt{2\psi_t} - \sqrt{\psi_t}}{3 - 2\sqrt{2}} - \sqrt{\frac{\omega_t}{3 - 2\sqrt{2}}}, \qquad \mathrm{CS}_t = m \max \left\{ 0, \frac{2(e^{\zeta_t} - 1)}{1 + e^{\zeta_t}} \right\},$$

and the realized volatility is

$$\mathrm{RV}_t = m \left( \sum_{i=0}^{119} r_{t-i}^2 \right)^{1/2}.$$

# Appendix B.1. Remaining decomposition moments

| | USDT | | | USDC | | |
|---|---|---|---|---|---|---|
| | pre | crisis | post | pre | crisis | post |
| SD $D$ | 2.17 | 30.4 | 8.46 | 4.12 | 327 | 10.4 |
| SD $\delta$ | 1.63 | 30.7 | 8.31 | 3.36 | 327 | 8.28 |
| SD $b$ | 1.80 | 4.08 | 3.78 | 5.21 | 38.3 | 8.83 |
| $\mathrm{Var}(b)/\mathrm{Var}(D)$ | 0.68 | 0.02 | 0.20 | 1.60 | 0.01 | 0.72 |
| $2\mathrm{Cov}(\delta, b)/\mathrm{Var}(D)$ | $-0.24$ | $-0.04$ | $-0.17$ | $-1.27$ | $-0.01$ | $-0.35$ |
| $\lvert b\rvert$ 95th percentile | 3.32 | 8.47 | 7.72 | 9.76 | 81.0 | 17.3 |

Remaining decomposition moments on synchronous minutes ($F_t = 1$), in bps.

# Appendix B.2. Control coefficients and fit

| | USDT | | | USDC | | |
|---|---|---|---|---|---|---|
| | pre | crisis | post | pre | crisis | post |
| $\theta$, CS of BTC/$Q$ | 0.001 | −0.011 | 0.273** | 0.041 | −1.895 | −0.695 |
| | (0.05) | (−0.11) | (2.01) | (0.40) | (−1.01) | (−1.05) |
| $\theta$, Kyle of BTC/$Q$ | 0.025 | 0.315* | −0.537** | −0.033 | 0.366 | −0.502* |
| | (0.70) | (1.79) | (−2.07) | (−0.26) | (0.25) | (−1.69) |
| $\theta$, Kyle of $Q$/USD | 0.016 | −0.106 | −0.075 | −0.220 | 5.517*** | −0.081 |
| | (0.58) | (−0.70) | (−0.44) | (−1.48) | (2.86) | (−0.14) |
| $\kappa$ | 0.158** | −0.056 | 0.075 | 0.298* | 0.989 | 1.442*** |
| | (2.54) | (−0.44) | (0.76) | (1.79) | (0.86) | (2.76) |
| $R^2$ | 0.408 | 0.290 | 0.342 | 0.442 | 0.373 | 0.378 |

Control coefficients and fit of the pass-through regression, with $t$ statistics in parentheses. *, ** and *** denote significance at 10%, 5% and 1%.

# Appendix C.1. Spillover index

Write the VAR in its moving-average form,

$$\mathbf{r}_t = \boldsymbol{\mu} + \sum_{h=0}^{\infty} \mathbf{A}_h \boldsymbol{\varepsilon}_{t-h}, \qquad \mathbf{A}_0 = \mathbf{I}, \qquad \mathbf{A}_h = \sum_{l=1}^{\min(h,p)} \boldsymbol{\Phi}_l \mathbf{A}_{h-l}.$$

The generalized forecast error variance decomposition uses only $\mathbf{A}_h$ (Pesaran and Shin, 1998),

$$\phi_{ij}(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} \left(\mathbf{e}_i^\top \mathbf{A}_h \boldsymbol{\Sigma} \mathbf{e}_j\right)^2}{\sum_{h=0}^{H-1} \mathbf{e}_i^\top \mathbf{A}_h \boldsymbol{\Sigma} \mathbf{A}_h^\top \mathbf{e}_i}, \qquad \tilde{\phi}_{ij} = \frac{100\,\phi_{ij}}{\sum_l \phi_{il}},$$

- $\sigma_{jj}$ is the $j$th diagonal element of $\boldsymbol{\Sigma}$, and $\mathbf{e}_i$ selects pair $i$.
- Each row is normalized so that $\sum_j \tilde{\phi}_{ij} = 100$, since the generalized shares of a row need not sum to 1.
- The total spillover index $S$ is the average share of forecast error variance that comes from the other four pairs (Diebold and Yilmaz, 2012).

# Appendix C.2. Granger causality

For Granger causality from pair $j$ to pair $i$, let $\gamma_{ij} = ([\mathbf{\Phi}_1]_{ij}, \ldots, [\mathbf{\Phi}_p]_{ij})^\top$ collect the $p$ coefficients of $j$ in the equation of $i$. The null hypothesis $H_0 \colon \gamma_{ij} = \mathbf{0}$ is tested with

$$W_{ij} = \hat{\gamma}_{ij}^\top \left[ \widehat{\mathrm{Var}} \left( \hat{\gamma}_{ij} \right) \right]^{-1} \hat{\gamma}_{ij} \ \sim \ \chi_p^2,$$

where $\widehat{\mathrm{Var}}(\hat{\gamma}_{ij})$ is the corresponding block of the estimated covariance of the VAR coefficients (Granger, 1969).